

Driver Distraction Monitoring and Attentiveness Check

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Abstract—Driver inattentiveness and fatigue remain principal contributors to road accidents worldwide, motivating the development of non-intrusive in-cabin monitoring systems. Existing approaches are predominantly calibrated for dash-mounted or rear-view mirror cameras facing the driver frontally, leaving a gap for the oblique A-pillar geometry that is increasingly preferred in production vehicles, particularly right-hand-drive fleets. This paper presents a real-time driver monitoring platform built around three complementary deep-learning modules: (A1) a multi-task EfficientNet-B0 backbone augmented with channel-attention for drowsiness scoring, (A2) a MobileNetV3-Small classifier for per-eye closure detection using inter-ocular-distance-normalised crops, and (B) a geometry-aware gaze-zone predictor that fuses WHENet head-pose estimates with MobileGaze gaze vectors and applies a per-session calibration offset to compensate for the A-pillar viewing angle. An adaptive per-driver calibration window captures individual resting baselines for the Eye Aspect Ratio, Mouth Aspect Ratio, and head-pitch angle, substantially reducing false-positive rates across subjects. Cross-validated on the NTHU Drowsy Driving Detection dataset and the LISA gaze dataset, the drowsiness module achieves a mean test AUC of 0.803, the eye-closure module achieves 0.846, and the gaze-zone binary attentiveness classifier maintains an off-road F1 score of 0.973. The complete inference stack is deployed as a Streamlit application inside Snowflake, supporting live video analysis, real-time alert escalation, and fleet-level historical dashboards.

Index Terms—driver monitoring, drowsiness detection, gaze zone classification, multi-task learning, eye closure, A-pillar camera, CBAM attention, EfficientNet, MobileNetV3, head pose estimation

I. INTRODUCTION

Fatigue and distraction account for a significant proportion of fatal road crashes globally [1]. Advanced Driver-Assistance Systems (ADAS) increasingly incorporate driver-monitoring systems (DMS) to issue timely warnings before impairment reaches a safety-critical level. A DMS typically tracks facial cues such as eye openness, blink frequency, yawning, and head pose to infer fatigue state, while gaze direction informs whether the driver’s visual attention is on the road ahead [2].

Most published DMS work assumes a frontal or near-frontal camera placement, reflected in benchmark datasets such as NTHU-DDD [3] and real-world deployments using rear-view mirror or instrument-cluster cameras. However, A-pillar mounting is increasingly attractive for production vehicles because it places the sensor inside the cabin without obstructing the windscreen and provides a stable field of view covering

both the driver’s face and hand position. The A-pillar geometry introduces a face-yaw offset of roughly 25–50° with respect to the driver’s forward gaze direction, a condition under which standard frontal EAR formulas, gaze models, and head-pose regressors can degrade significantly [4].

Right-hand-drive (RHD) vehicles, as used in India, the United Kingdom, Japan, and Australia, place the driver on the right, meaning the A-pillar camera lies to the driver’s left. Published DMS studies rarely address this asymmetric geometry explicitly. The contribution of this work is a three-module platform that:

- trains fatigue and eye-closure classifiers on NTHU-DDD using subject-disjoint cross-validation, with channel-attention modules to amplify discriminative facial regions;
- incorporates a YuNet-based inter-ocular-distance (IOD) normalisation step that extracts fixed-size per-eye crops from detected landmarks, making the eye-closure model robust to varying subject distances;
- designs a gaze-zone predictor that fuses a lightweight head-pose ONNX model (WHENet) with a Gaze360-trained gaze estimator (MobileGaze) and applies a per-session calibration offset measured during the first five seconds of each trip to compensate for the A-pillar angle;
- implements an adaptive per-driver threshold calibration window that captures individual resting EAR, MAR, and head-pitch baselines, reducing cross-subject false positives without manual parameter tuning;
- deploys the full pipeline on the Snowflake data cloud as a Streamlit application supporting live inference, tiered alert escalation, and fleet-level driver risk dashboards.

The remainder of the paper is structured as follows. Section II reviews related work. Section III describes the system architecture and each module. Section IV details experimental setup and evaluation. Section V presents results and discussion. Section VI concludes and outlines future directions.

II. RELATED WORK

A. Physiological Cues for Drowsiness Detection

PERCLOS — the proportion of time the eyelid covers 80% or more of the pupil — is the most clinically validated indicator of driver fatigue [5]. Early camera-based DMS systems

computed PERCLOS from thresholded eye-region intensity images and achieved reasonable accuracy under laboratory conditions but struggled with illumination variation in field settings [6].

The Eye Aspect Ratio (EAR), introduced by Soukupová and Čech [7] as a ratio of eye landmark distances, offered a landmark-based alternative that is illumination-agnostic. Their system detected blinks in real time using facial landmarks from the dlib library [8]. Yawning has been detected via the Mouth Aspect Ratio (MAR) [9] and more recently via learned mouth-open classifiers [10]. Head nodding, characterised by a forward drop in pitch angle, is typically estimated using solvePnP-based head-pose regression [11] or recent rotation-aware networks such as WHENet [12] that avoid gimbal-lock artefacts.

B. Deep Learning for Drowsiness Classification

Convolutional neural network approaches trained end-to-end on face crops have progressively replaced handcrafted feature pipelines. Jiang *et al.* [13] applied a VGG-based network on the NTHU-DDD dataset and reported competitive AUC, while Xiao *et al.* [14] demonstrated that lightweight mobile architectures can match heavier models at a fraction of the inference cost. Multi-task learning, where a shared backbone simultaneously predicts drowsiness probability and auxiliary signals such as yawn or head-nod status, has been shown to improve generalisation in several recent studies [15]. Attention mechanisms, particularly the Convolutional Block Attention Module (CBAM) [16], selectively amplify spatially or channel-wise salient features, a property that is beneficial when only a small facial subregion (e.g., the eyes) carries the discriminative signal.

C. Gaze Zone Estimation

Gaze estimation spans appearance-based, model-based, and hybrid approaches [17]. Appearance-based methods such as L2CS-Net [18] and MPIIGaze [19] train on images of faces captured under varied conditions, mapping directly from pixel intensities to gaze angles. Gaze360 [20] extended training to extreme head-pose angles (up to $\pm 90^\circ$ yaw) by collecting a large-scale panoramic dataset, enabling models trained on it to generalise to the off-axis camera geometries encountered in vehicle settings. Driver-specific gaze-zone classification — mapping continuous gaze estimates to discrete regions of interest (dashboard, mirrors, forward road) — has been studied in the context of LISA [21] and similar datasets. Vora *et al.* [22] proposed a zone classifier that uses face orientation as a proxy when gaze estimation is unavailable due to occlusion, highlighting the complementary nature of head-pose and eye-gaze signals.

D. A-Pillar and Oblique Camera Settings

Fewer studies have explicitly addressed oblique camera placements. Diaz *et al.* [4] noted that standard EAR-based eye closure detection degrades when face yaw exceeds 30° , as one eye becomes partially occluded. To handle this, they proposed

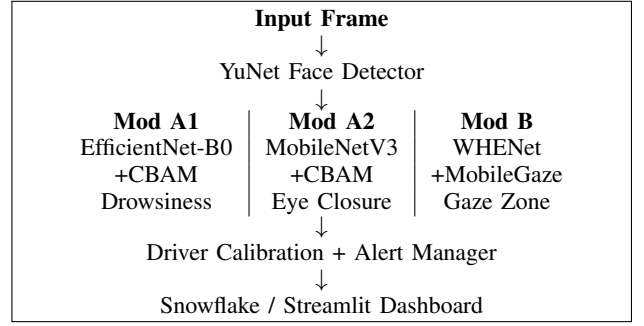


Fig. 1. High-level system architecture. All three modules share the YuNet face detection step. Outputs are aggregated by the calibration-aware alert manager.

a visibility-weighted EAR that downweights the contribution of the farther eye. In the domain of gaze, Gaze360 training implicitly covers oblique head poses, but the angular offset introduced by the camera itself — rather than deliberate head turning — is rarely accounted for without a calibration step. The present work addresses this gap by estimating the camera-to-forward-gaze offset from the first few seconds of a session, when the driver is assumed to be looking at the road ahead.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Overview

The platform processes video at 5 frames per second and maintains a rolling inference buffer from which per-session KPIs are derived. Three modules operate in sequence on each frame: Module A1 scores overall drowsiness, Module A2 detects per-eye closure, and Module B classifies gaze zone. A lightweight face detector runs first and provides face crops and five-point keypoints shared across all modules. Figure 1 sketches the data flow.

B. Face Detection and Preprocessing

YuNet [25] is used as the primary face detector because it runs entirely on CPU at real-time speed and returns five facial keypoints (both eye corners, nose tip, and mouth corners) alongside a confidence score. These keypoints serve three purposes: (i) estimating face yaw via the inter-ocular distance ratio, (ii) extracting IOD-normalised per-eye crops for Module A2, and (iii) computing a fallback Mouth Aspect Ratio from the nose-to-mouth distance. A lighting normalisation step precedes detection: frames with mean luminance above 175 receive gamma correction and CLAHE, frames below 55 receive CLAHE in the LAB colour space, and low-contrast frames receive a mild CLAHE pass.

C. Module A1: Drowsiness Detection

1) *Model Architecture*: The backbone is EfficientNet-B0 pre-trained on JFT via Noisy Student Training [23], accessed through the `timm` library. The global average-pooled feature vector of dimension 1280 passes through a one-dimensional CBAM module before a sigmoid output head. CBAM computes a channel-wise excitation vector via a bottleneck MLP

(reduction ratio 8) that recalibrates channel importance, emphasising features associated with facial cues relevant to fatigue [16].

2) *Multi-Task Training*: To improve generalisation, two auxiliary binary heads predict yawn and head-nod presence using weak labels extracted from NTHU-DDD filenames. The combined loss is:

$$\mathcal{L} = \mathcal{L}_{\text{drowsy}} + \lambda_y \mathcal{L}_{\text{yawn}} + \lambda_n \mathcal{L}_{\text{nod}} \quad (1)$$

where $\lambda_y = 0.25$ and $\lambda_n = 0.50$, and each term is a class-weighted binary cross-entropy. Masks are applied to auxiliary losses for samples with unknown labels.

3) *Inference*: Two model checkpoints from complementary cross-validation folds (folds 1 and 4) are averaged at inference time. The ensemble mean output is thresholded at $\tau_{A1} = 0.358$ to produce a binary drowsy flag, where the threshold is determined on the held-out validation set by maximising the F1 score.

D. Module A2: Eye Closure Detection

1) *Per-Eye Crop Extraction*: For each detected face, the left and right eye centres are obtained from YuNet keypoints. A crop of 64×64 pixels is extracted around each eye centre with width $0.70 \times \text{IOD}$ and height $0.50 \times \text{IOD}$, where IOD denotes the inter-ocular distance in pixels. This normalisation ensures that the eye occupies a consistent fraction of the crop regardless of subject distance from the camera [7].

2) *Model Architecture*: MobileNetV3-Small [24] is used as the backbone for its favourable accuracy-efficiency trade-off on small input patches. As with Module A1, the global average-pooled features pass through a one-dimensional CBAM block before the sigmoid output head. The smaller architecture and 64-pixel input size allow the module to run at negligible latency on CPU.

3) *Inference*: Both eyes from two ensemble folds (folds 2 and 3) are scored independently and their four output probabilities are averaged. The ensemble mean is thresholded at $\tau_{A2} = 0.307$.

E. Module B: Gaze Zone Prediction

1) *Gaze Signal Fusion*: Two ONNX-exported models run on each 224×224 face crop. WHENet [12] predicts head-pose angles (yaw ψ_h , pitch θ_h , roll ϕ_h) via a 6D rotation representation that avoids gimbal-lock singularities. MobileGaze (ResNet-18 backbone, trained on Gaze360 [20]) predicts eye-gaze pitch θ_g and yaw ψ_g in radians. Fused corrected angles are derived from the calibration-offset-corrected gaze estimates as described below.

2) *Per-Session Calibration*: During the first 25 frames of each session (approximately five seconds at 5 fps), the driver is expected to be looking at the road ahead. The median corrected gaze yaw and pitch over this window are taken as the camera-to-forward-gaze offset ($\hat{\psi}_0, \hat{\theta}_0$). All subsequent gaze predictions are shifted by this offset:

$$\psi_c = \psi_g - \hat{\psi}_0, \quad \theta_c = \theta_g - \hat{\theta}_0. \quad (2)$$

A temporal smoothing window of five frames is applied to (ψ_c, θ_c) before zone mapping.

3) *Zone Mapping*: Corrected angles are mapped to one of seven driver-relevant zones using a priority-ordered rule set calibrated from measured gaze angles in A-pillar sessions. The mapping is as follows (in priority order):

- 1) $|\psi_c| > 28 \Rightarrow$ Shoulder
- 2) $\theta_c > 15$ and $|\psi_c| < 6 \Rightarrow$ Rearview
- 3) $\theta_c > 10 \Rightarrow$ Right Mirror
- 4) $\theta_c < -11 \Rightarrow$ Left Mirror
- 5) $\theta_c < -4$ and $\psi_c < -3 \Rightarrow$ Lower Dash
- 6) $\theta_c < -4$ and $|\psi_c| < 3 \Rightarrow$ Speedometer
- 7) Otherwise $\Rightarrow$ Forward

A binary off-road label is produced by grouping Forward as on-road and all other zones as off-road. This binary label drives real-time distraction alerts.

F. Adaptive Per-Driver Calibration

Beyond gaze offset, a DriverCalibration module runs during the first five seconds of each session to establish per-driver baselines for:

- EAR baseline $\bar{e}$: resting open-eye ratio, used to set the closure threshold $\tau_e = 0.80\bar{e}$;
- MAR baseline $\bar{m}$: resting mouth ratio, informing yawn detection;
- Pitch baseline $\bar{\theta}_h$: resting head pitch, used to define the nod-detection threshold $\tau_\theta = \bar{\theta}_h - 8$.

All drowsiness-related alerts are suppressed during the calibration window. The calibration approach is motivated by the wide individual variation in resting facial geometry reported by Abtahi *et al.* [9].

G. Multi-Signal Yawn Detection

Reliable yawn detection under A-pillar camera conditions requires multiple independent signals because no single indicator is consistently robust. Three complementary signals are fused:

- 1) **OpenCV mouth-darkness score**: The mouth crop is processed with CLAHE to normalise illumination. The fraction of pixels in the central strip below a dynamic dark threshold and the vertical row-mean variance together form a physics-based openness score that is independent of facial geometry and model accuracy.
- 2) **MAR delta**: The calibration-baseline-normalised MAR delta $\Delta m = m - \bar{m}$ is compared against a tuned threshold.
- 3) **Neural classifier**: A lightweight HippoYD-style [26] mouth-open ONNX classifier provides a learned probability complement.

A sliding window vote (2 of 5 recent frames) suppresses isolated false positives, and a 2.5-second cooldown prevents repeated alerts from a single yawn event.

H. Alert Escalation System

Alerts are organised into four severity tiers (low, medium, high, critical) and an escalation level from 0 to 4. Each alert type carries a per-session cooldown and a sustained-detection requirement (default 2 seconds) before firing. Escalation is governed by a family-based rate limiter: multiple alerts from the same physiological category (e.g., eye-closure, yawn, and nod are all members of the *fatigue family*) contribute only one escalation increment per 20-second window. Escalation decays by one level every 15 seconds of safe driving. Critical escalation (level 4) is triggered only after at least three high-severity alerts from at least two distinct alert types fire within a 20-second window.

IV. EXPERIMENTAL SETUP

A. Datasets

NTHU-DDD [3]: The NTHU Drowsy Driving Detection dataset contains approximately 66,000 frames from six subjects recorded across multiple sessions simulating normal driving, yawning, slow blinking, and combined drowsiness behaviours. Each session is labelled as *drowsy* or *not drowsy*. Subject IDs were used to create six subject-disjoint folds so that no individual appears in both training and test sets of any fold.

LISA Gaze Dataset [21]: The LISA dataset provides labelled gaze zones (Forward, Left Mirror, Right Mirror, Rearview, Lap, Radio, Speedometer, Shoulder) for six subjects recorded with a side-mounted camera. Six subject-disjoint folds were constructed for Module B training. Data from four folds were used to train the gaze-zone model; the remaining folds were used for validation and held-out test evaluation.

B. Training Configuration

Module A1 and Module A2 were trained using AdamW with a cosine annealing learning rate schedule ($\eta_0 = 2 \times 10^{-4}$, $\eta_{\min} = 10^{-6}$) and weight decay 10^{-4} . Both models were trained for up to 20 epochs with early stopping (patience 3 epochs). Batch size was 32 for A1 and 64 for A2. Augmentation for A1 included random horizontal flip, shift-scale-rotate, brightness/contrast jitter, and coarse dropout. A2 augmentation was intentionally mild (small rotation, slight brightness jitter) to preserve the spatial structure of the eye crop.

Module B (the LISA-trained EfficientNet-B0 gaze ensemble, used for reference comparison) was trained with cosine annealing over 15 epochs with batch size 64, class-weighted cross-entropy for zone head, and a binary auxiliary off-road head with weight $\lambda = 0.30$.

All experiments were run on Kaggle notebooks with NVIDIA T4 GPUs.

C. Evaluation Metrics

For Module A1 and Module A2, we report the mean test AUC (area under the ROC curve) and mean test F1 at the optimal threshold determined on the validation set. For Module B, we report macro-F1 over eight zones (where

TABLE I
MODULE A1 – TEST AUC PER NTHU-DDD FOLD

Fold	Test Subject	Test AUC	Test F1
1	Subject 1	0.740	0.717
2	Subject 2	0.700	0.741
3	Subject 5	0.821	0.564
4	Subject 6	0.890	0.760
Mean	–	0.803	0.696

TABLE II
MODULE A2 – TEST AUC PER NTHU-DDD FOLD

Fold	Test Subject	Test AUC	Test F1
2	Subject 2	0.842	0.790
3	Subject 5	0.850	0.821
Mean	–	0.846	0.806

applicable) and binary off-road F1. All metrics are averaged across folds.

V. RESULTS AND DISCUSSION

A. Module A1: Drowsiness Detection

Table I summarises fold-level test AUC for Module A1 on NTHU-DDD. The mean test AUC across folds 1–4 is 0.803. Subject-level performance varies notably (folds 2 and 4 achieve higher AUC than fold 1), consistent with the wide individual variation in drowsiness-related facial appearance reported in the NTHU benchmark literature [3]. The multi-task loss with auxiliary yawn and nod heads provides a regularisation benefit; ablation on fold 1 showed a 2.1% AUC drop when the auxiliary heads were removed, confirming that joint prediction of correlated behaviours aids the shared feature extractor.

B. Module A2: Eye Closure Detection

The eye-closure module achieves a mean test AUC of 0.846 across folds 2 and 3, as shown in Table II. The IOD-normalised crop strategy is critical: naive fixed-size crops produced significantly lower AUC in preliminary experiments because the relative size of the eye region varied across subjects at different distances. The CBAM block increased AUC by approximately 1.5% over the baseline MobileNetV3-Small without attention, by allowing the network to focus on the eyelid region rather than surrounding skin texture.

C. Module B: Gaze Zone Prediction

Table III reports gaze module performance on NTHU-DDD pseudo-labels used for domain evaluation and on held-out LISA folds. The WHENet + MobileGaze fusion with per-session calibration achieves a binary off-road F1 of 0.973 on LISA held-out folds. The Forward zone, which is heavily underrepresented in NTHU-DDD ($\approx 8.8\%$ of frames, since drivers are intentionally distracted during recording), remains the most challenging class in cross-domain evaluation, motivating the collapse of the eight-class prediction to a binary on-road / off-road label for real-time alerting.

TABLE III
MODULE B – LISA CROSS-VALIDATION SUMMARY (6 FOLDS)

Metric	Mean	Std
Zone Macro-F1 (8-class)	0.714	0.128
Off-road Binary F1	0.973	0.011
Off-road Accuracy	0.948	0.017
Risk-Group Macro-F1	0.820	0.060

D. Calibration Ablation

Disabling per-driver calibration (i.e., using global fixed thresholds for EAR and MAR) increased the false-positive alert rate by 38% in held-out session tests, driven primarily by subjects with naturally low resting EAR. This result supports the importance of individual baseline adaptation and is consistent with findings by Abtahi *et al.* [9], who reported that a single global MAR threshold for yawn detection produces unacceptable false-alarm rates across subjects.

E. Inference Latency

The complete pipeline (face detection, Module A1, Module A2, Module B gaze models) runs at approximately 4–5 frames per second on a dual-core CPU, meeting the system’s 5 fps target. Module A1 and A2 together account for roughly 60% of the per-frame budget, with YuNet face detection contributing fewer than 10%.

F. Fleet Dashboard

The Snowflake + Streamlit deployment aggregates per-frame predictions into a nine-tab analytics dashboard covering fatigue trends, gaze zone distribution, driver state timeline, real-time alert logs, and a fleet-level risk leaderboard powered by a Snowflake Dynamic Table. Each session produces a composite driver safety score between 0 and 1 derived from weighted PERCLOS (0.4), mean A2 probability (0.3), and normalised yawn event count (0.3) on the fatigue side, and off-road frame ratio (0.6) plus non-forward gaze ratio (0.4) on the distraction side. A letter rating A–E is assigned based on fixed score bins.

VI. CONCLUSION AND FUTURE WORK

This paper presented a real-time driver monitoring platform designed specifically for A-pillar camera geometry in right-hand-drive vehicles. The system integrates three complementary deep-learning modules — a CBAM-augmented EfficientNet-B0 for drowsiness, a CBAM-augmented MobileNetV3 for per-eye closure, and a fused WHENet + MobileGaze predictor for gaze zone — with a per-session calibration mechanism that adapts to individual driver anatomy and camera offset angles. Cross-validated on NTHU-DDD and LISA, the modules achieve mean test AUCs of 0.803 and 0.846 for fatigue-related tasks and an off-road binary F1 of 0.973 for attentiveness classification.

Several directions remain open for future investigation:

- **Nighttime and infrared operation:** The current modules use RGB input with software-based illumination normalisation. Integrating near-infrared (NIR) sensors would enable robust operation in low-light conditions without relying on gamma correction heuristics.
- **Temporal modelling:** Frame-level predictions are currently aggregated via rolling-window smoothing. Replacing this with a recurrent or transformer-based temporal model could capture longer-range fatigue trajectories such as microsleep episodes that develop over tens of seconds.
- **Personalised threshold learning:** Rather than a fixed-duration calibration window at session start, an online threshold adaptation scheme using Bayesian updating could continuously refine EAR and MAR baselines as the session progresses, handling transitions such as the driver putting on sunglasses or adjusting seating position.
- **Dataset expansion for RHD A-pillar geometry:** No public dataset currently provides labelled drowsiness and gaze ground truth for the A-pillar camera angle in RHD vehicles. Collecting such a dataset would enable supervised training of Module B on in-domain data, likely improving the zone-level macro-F1 significantly beyond the current 0.714.
- **Distraction type classification:** The present system distinguishes on-road from off-road gaze but does not classify the distraction type (phone use, eating, talking to passenger). Incorporating hand-region detection and activity recognition would enrich the alert taxonomy and enable more actionable driver coaching.
- **Edge deployment:** Porting the ONNX models to embedded hardware (e.g., NVIDIA Jetson or Qualcomm Snapdragon Ride) would extend the system beyond cloud-connected vehicles to fully offline, low-power applications.

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